

ACML101: Introduction to Chemistry

Inorganic Chemistry: Tutorial 3

(1) Which of the following complexes follow 18 electron rules

(a) $[\text{Cp}_2\text{Co}]$ (b) $[\text{Cp}_2\text{Co}]^+$ (c) $\text{V}(\text{CO})_6$ (d) $[\text{V}(\text{CO})_6]^-$ (e) $\text{MeMn}(\text{CO})_5$ (f) $(\eta^3\text{-C}_3\text{H}_5)_2\text{Rh}(\mu\text{-Cl})_2\text{Rh}(\eta^3\text{-C}_3\text{H}_5)_2$

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg

- (a) 19 electron
- (b) 18 electron
- (c) 17 electron
- (d) 18 electron
- (e) 18 electron
- (f) 18 electron

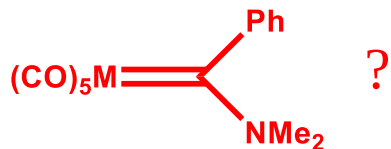
(2) Find the unknown quantity, if the central metal ion follows 18 electron rule.

(a) $\text{Re}(\text{CO})_n\text{Br}$ find n?

$$7 + (n \times 2) + 1 = 18; n = 5$$

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
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(b) Find M (1st row) in



$$(5 \times 2) + X + 2 = 18; x = 6; M = \text{Cr}$$

(c) $[\text{Os}(\text{CO})_4]^z$ find z?

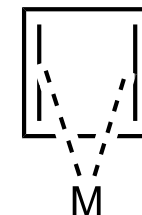
$8 + 4 \times 2 \pm x = 18; x = 2$; as the Os counts two electron from the charge, therefore $Z = -2$

(d) $(\eta^x\text{-C}_7\text{H}_8)\text{Mo}(\text{CO})_3$ find x?

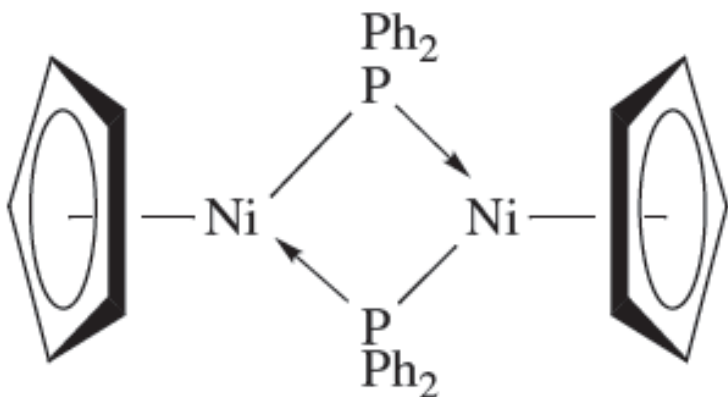
$$X + 6 + (3 \times 2) = 18; x = 6$$

(e) $[(\text{Cp})\text{Co}(\text{C}_n\text{H}_n)]$

$$5 + 9 + x = 18; x = 4; \text{C}_n\text{H}_n \text{ should be 4 electron donor, e.g. } = \text{C}_4\text{H}_4$$



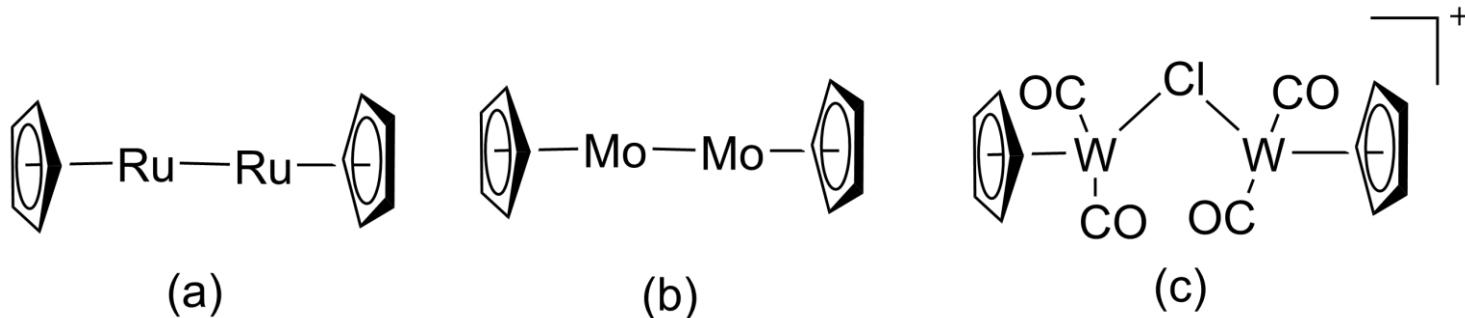
(3) Solely on the basis of the 18-electron rule, suggest whether $(\eta^5\text{-Cp})\text{Ni}(\mu\text{-PPh}_2)_2\text{Ni}(\eta^5\text{-Cp})$ might be expected to contain a metal–metal bond ($\text{Ni}^0 = [\text{Ar}]3\text{d}^84\text{s}^2$).



Ni(0) (group 10) contributes 10 electrons; $\eta^5\text{-Cp}$ gives 5 electrons. Two $\mu\text{-PPh}_2$ contribute 6 electrons, 3 per Ni. Total per Ni = 18 electrons.

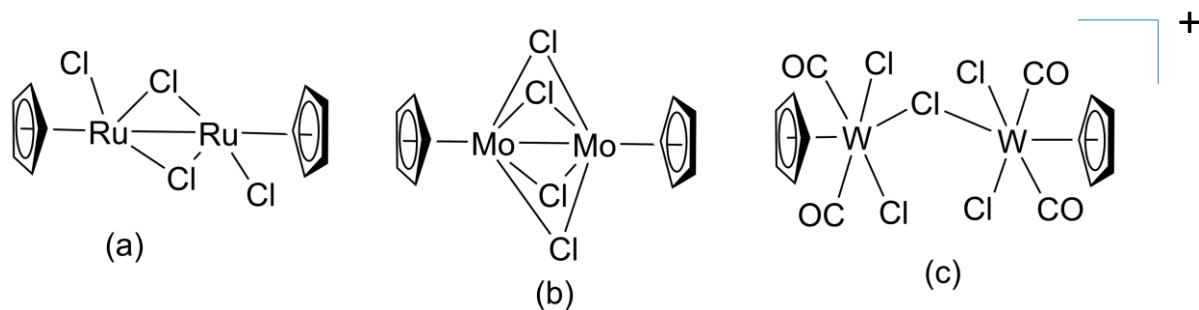
As both the Ni obeys the 18-electron rule and no Ni-Ni bond is required.

(4) Four chlorine ligands are missing in each of the given skeletons of dimeric compounds a, b and c. Given that all of them obey the 18-electron rule and no additional metal-metal bonds are present, attach the missing Cl ligands on the complexes in the appropriate manner.

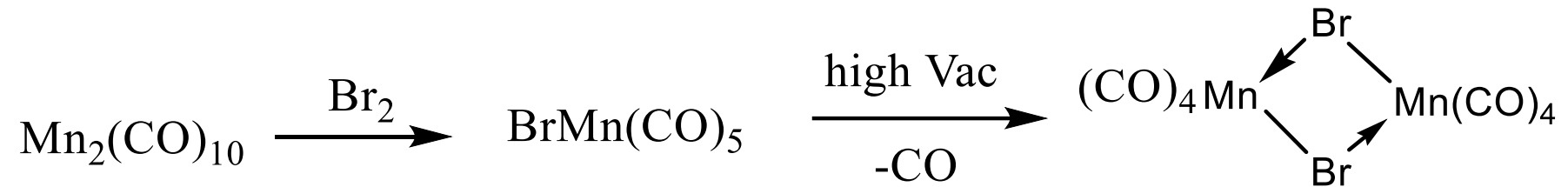


21	22	23	24	25	26	27	28	29	30
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La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg

Answer:



(5) Reaction of $\text{Mn}_2(\text{CO})_{10}$ with Br_2 resulted in an orange colored powdery compound A. On keeping A under high vacuum in a sealed tube at room temperature for several days resulted in the formation of dark orange crystals of a new compound B. Analysis of A and B indicated only infra red bands in the range of $2045\text{-}1950\text{ cm}^{-1}$. A and B are stable compounds and both contain bromine atoms. Draw the structures of A and B.



(6) Which case of $\text{MoL}_3(\text{CO})_3$ is expected to exhibit a lower CO stretching frequency; $\text{L} = \text{PF}_3$, or $\text{L} = \text{PMe}_3$? Explain

PMe_3 is a stronger σ -donor and weak π -acceptor compared to PF_3 . (as a consequence of the highly electronegative fluorines). As a result, π -back donation from the metal-to-CO is less in $[\text{Mo}(\text{CO})_3(\text{PF}_3)_3]$ compared to $[\text{Mo}(\text{CO})_3(\text{PMe}_3)_3]$, which results lower C–O stretching frequencies in $\text{Mo}(\text{CO})_3(\text{PMe}_3)_3$.

(7) Arrange the following in accordance with decreasing carbonyl stretching frequency ($\nu_{\text{C-O}}$) in cm^{-1} , $\text{Ni}(\text{CO})_4$, $[\text{Co}(\text{CO})_4]^-$ and $[\text{Fe}(\text{CO})_4]^{2-}$

All are 18 electron metal complex. With increasing the negative charge, back bonding from metal to the π^* orbital increases.

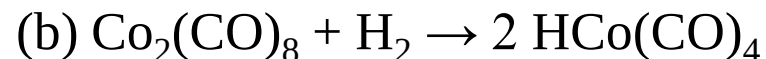


(8) Classify the following reactions as oxidative addition, reductive elimination, migratory insertion, β -hydrogen elimination, ligand coordination change or simple addition. (Check the change of oxidation state and coordination number)

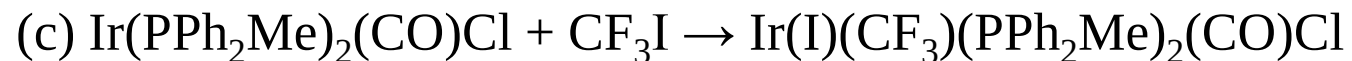
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Migratory insertion



Oxidative addition

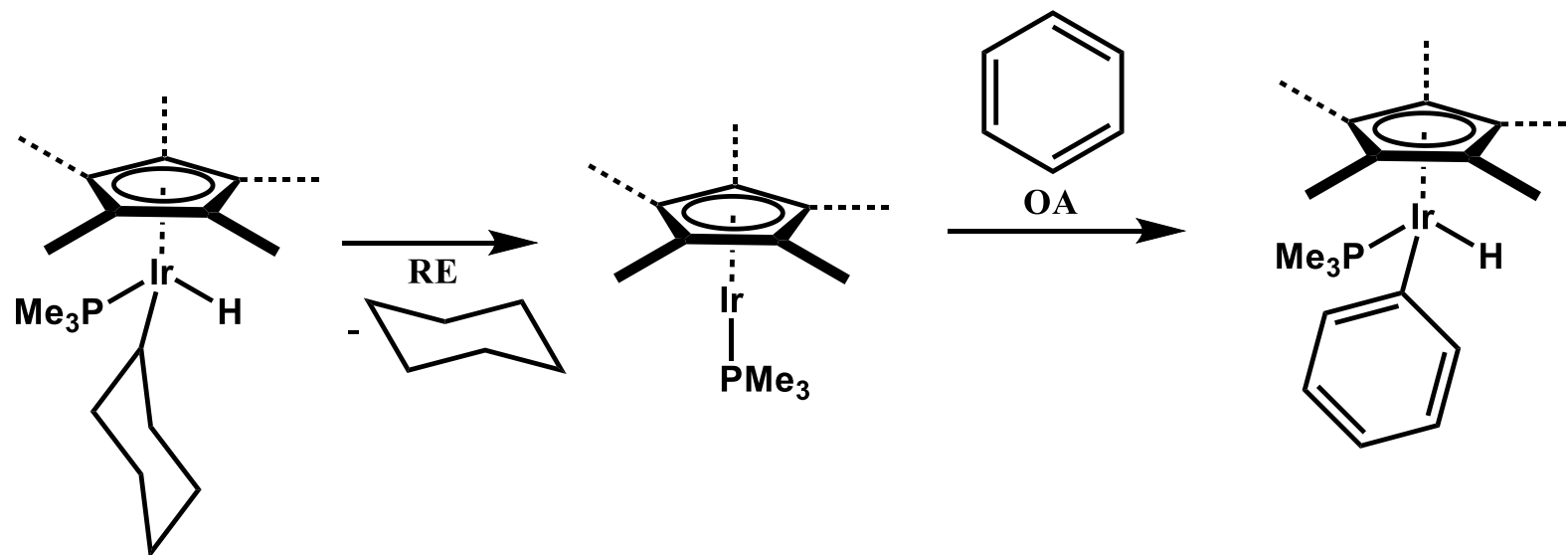
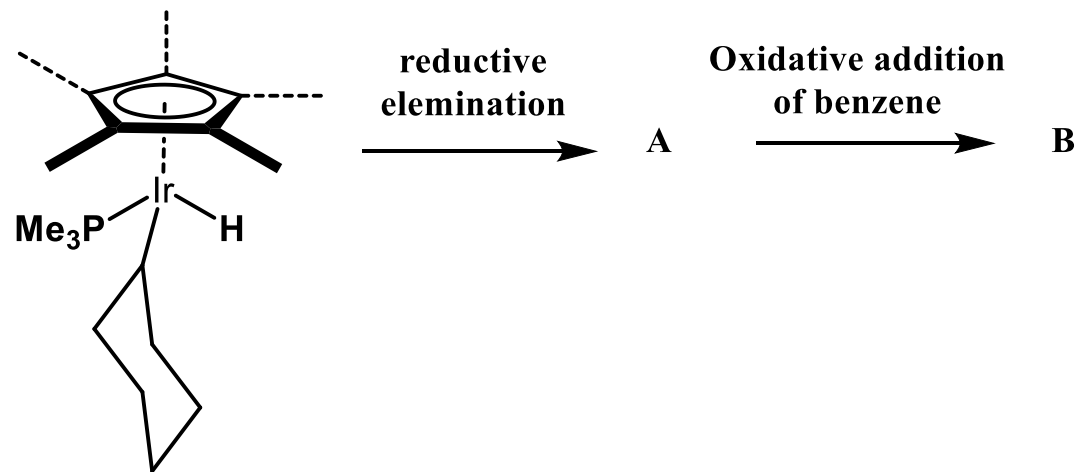


Oxidative addition



Simple addition

(9) Predict the product of the following reaction? Count the change of oxidation state and valence shell electrons of the metal center. (Ir = [Xe]5d⁷6s²)(previous year question)



10. The complex $(\text{PPh}_3)_3\text{Rh}(\text{Me})$ is cleaved in the presence of D_2 and forms methane and $(\text{PPh}_3)_2\text{Rh}(\text{D})(\text{PPh}_2\text{C}_6\text{H}_4\text{D})$. Suggest a suitable mechanism for the reaction. (Hints: 1st step is intramolecular oxidative addition)

Answer

